NeoSmith: System Validation & Inference Logs

Version: 2.0.0 | **Nature:** Empirical Agentic Inference Log

The system is not just code; it is a reflection of reality. These logs represent the agent's ability to parse complex organizational structures and deliver the truth behind the network.

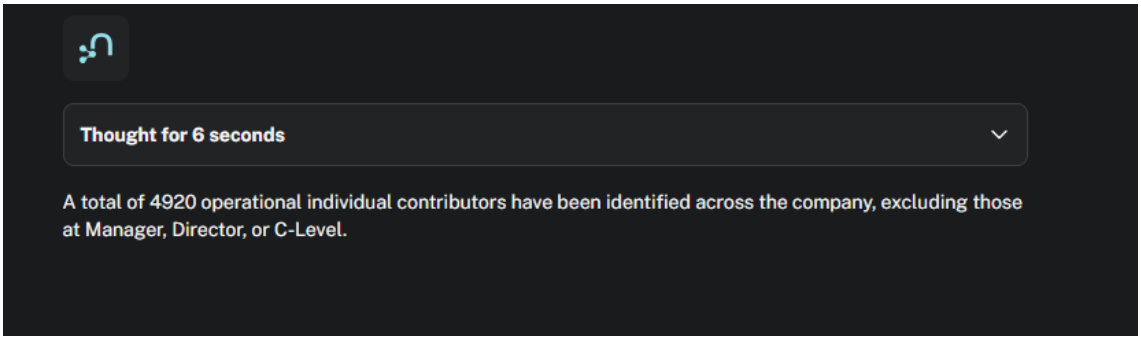
1. Overview

This document provides a set of real-world inference samples generated by the NeoSmith Agentic GraphRAG system. These logs validate the agent's capability to execute complex organizational queries—from ONA (Organizational Network Analysis) to talent reallocation—while adhering to the cognitive guardrails and semantic constraints defined in our PRD and UAT Playbook.

2. Inference Samples

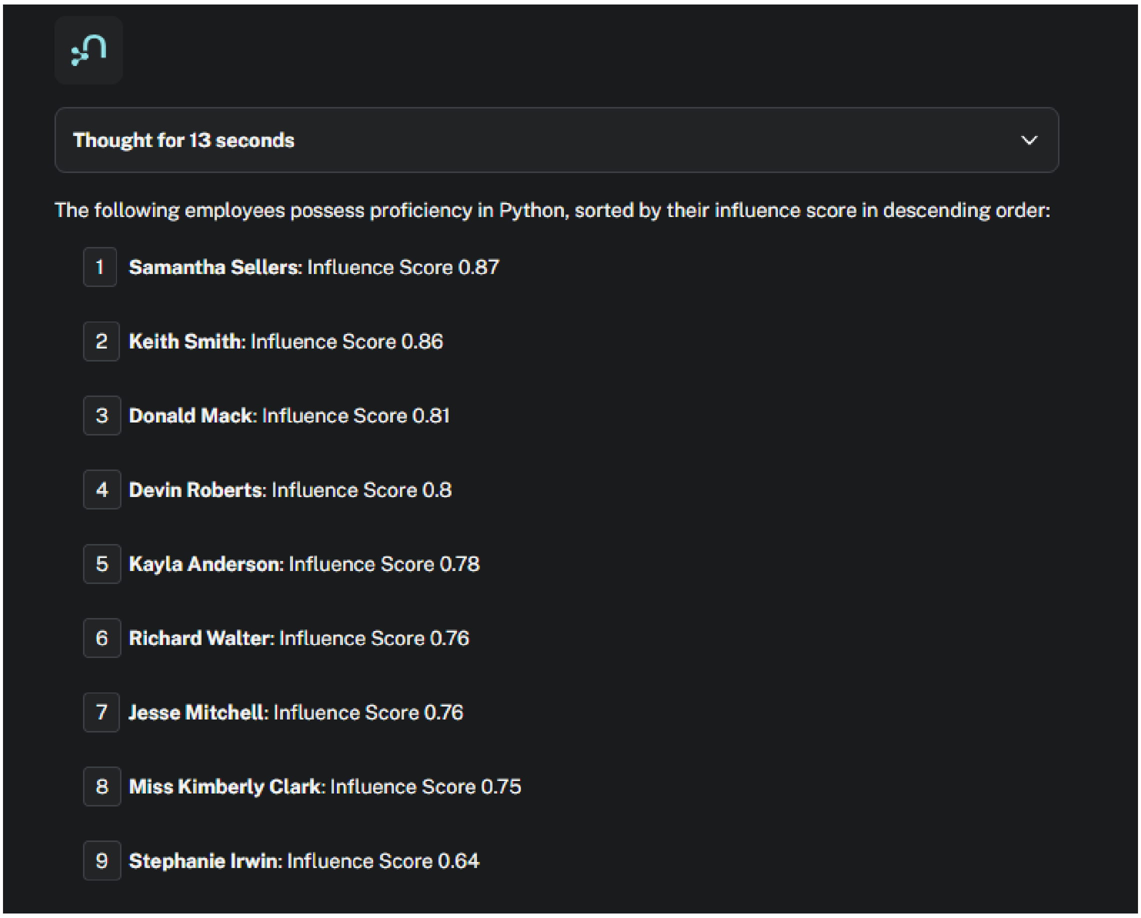
Sample 01: Administrative Aggregation (Individual Contributors)

Prompt: "Give me a total count of all operational individual contributors across the entire company. Exclude anyone at the Manager, Director, or C-Level."



Sample 02: Skill-Based Sourcing (Python Experts)

Prompt: "Identify all employees who possess proficiency in Python. I need them sorted so that the most influential figures within our informal network appear first."



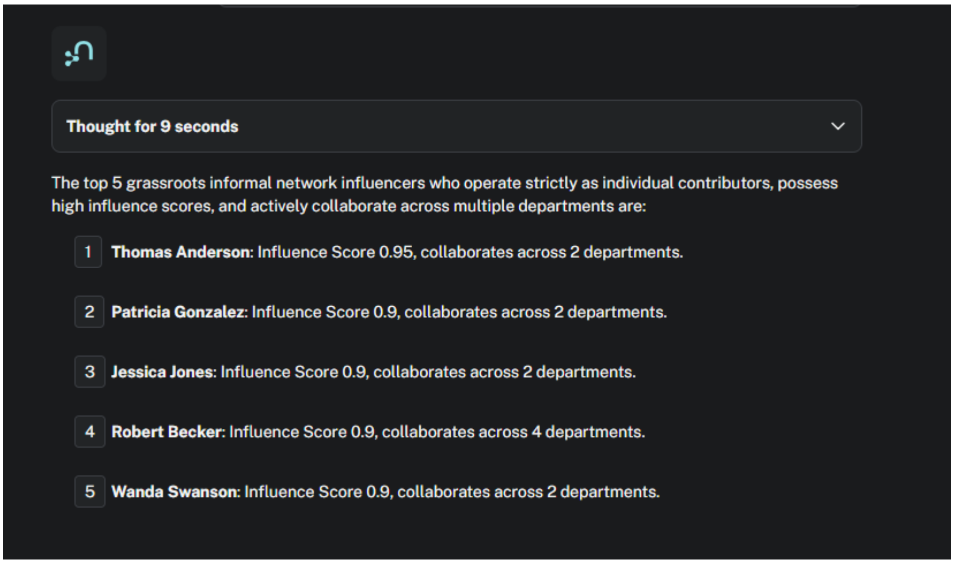
Thought for 13 seconds

The following employees possess proficiency in Python, sorted by their influence score in descending order:

1. **Samantha Sellers:** Influence Score 0.87
2. **Keith Smith**: Influence Score 0.86
3. **Donald Mack**: Influence Score 0.81
4. **Devin Roberts**: Influence Score 0.8
5. **Kayla Anderson:** Influence Score 0.78
6. **Richard Walter:** Influence Score 0.76
7. **Jesse Mitchell**: Influence Score 0.76
8. **Miss Kimberly Clark**: Influence Score 0.75
9. **Stephanie Irwin**: Influence Score 0.64

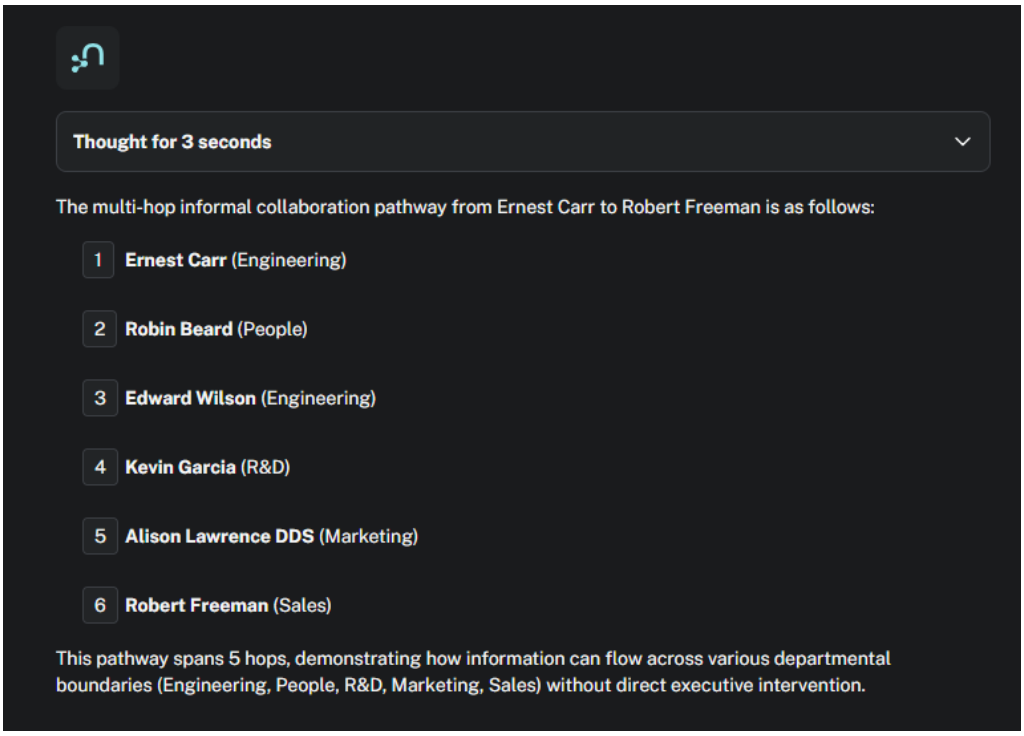
Sample 03: Grassroots Influencer Mapping

Prompt: " Identify the top 5 grassroots informal network influencers who operate strictly as individual contributors. Look at the role level property and exclude any nodes where the level is 'Manager', 'Director', or 'C-Level'. They must have high influence scores and actively collaborate across multiple departments."



Sample 04: Multi-Hop Collaboration Pathway

Prompt: " Trace the multi-hop informal collaboration pathway from Ernest Carr to Robert Freeman to evaluate how information flows across boundaries without executive intervention "



Sample 05: Talent Circularity Matrix (Finance Department)

Prompt: " We are executing a full-scale operational transition for the department named Finance. Generate a comprehensive talent circularity matrix matching all affected team members to all compatible open positions company-wide. Do not truncate the result to a single candidate; map out all viable placement options sorted by minimal training weeks required."

Thought for 8 seconds

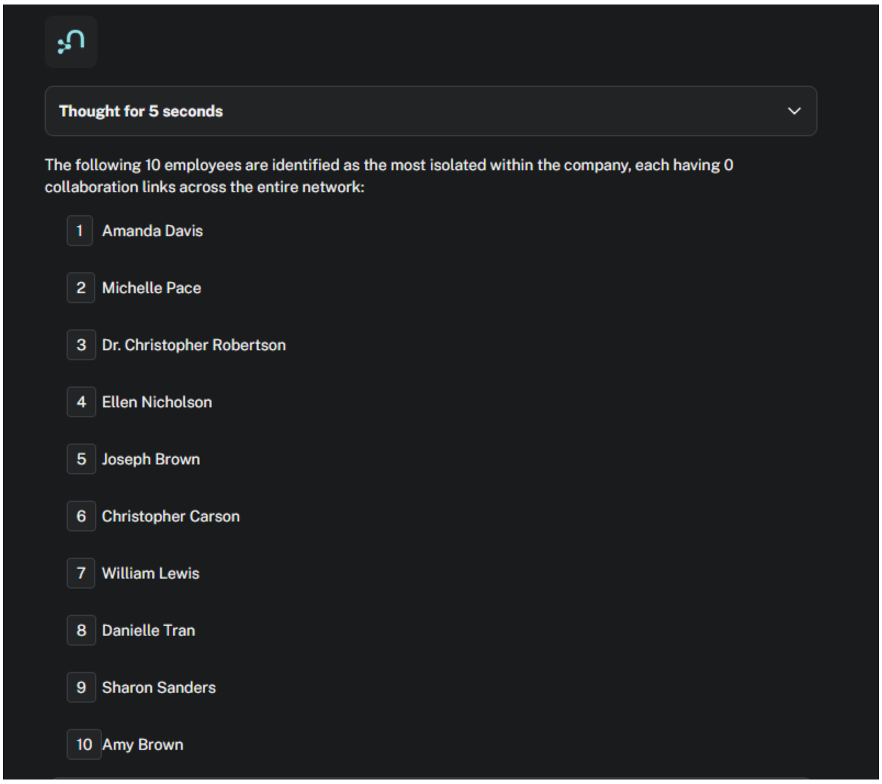
A comprehensive talent circularity matrix for the Finance department has been generated, matching all affected team members to compatible open positions company-wide. The results are sorted by minimal training weeks required.

Here are the viable placement options:

- **Patricia Gonzalez** (Finance)
  - **Current Skills:** Critical Thinking, Public Speaking, Asset Management, Emotional Intelligence, Design Thinking
  - **Matching Required Skills:** Brand Strategy, SEO/SEM, Growth Hacking
  - **Position Title**: Marketing Specialist
  - **Target Department:** Marketing
  - **Hiring Manager**: Leia Organa
  - **Training Weeks Required:** 0 (for all listed skill matches)

Sample 06: Structural Silo Detection

Prompt: "Identify organizational structural silos by listing the top 10 employees who are most isolated inside the company, meaning they have the lowest number of total collaboration links across the entire network."



Sample 07: Targeted Individual Career Trajectory Analysis (Jake Peralta)

Prompt: "Execute a targeted talent mobility routing analysis specifically for Jake Peralta. Map his skills against global open positions and check if there are placements available that require reskilling weeks."

Thought for 6 seconds

A talent mobility routing analysis has been performed for Jake Peralta, mapping his skills against global open positions.

The analysis indicates that Jake Peralta, currently in the Engineering Department, has skills such as Design Thinking, Emotional Intelligence, and Robotics that could be relevant to a "Marketing Specialist" role in the Marketing Department. The hiring manager for this role is Leia Organa.

For the identified matches, the required skills for the Marketing Specialist position include Growth Hacking, SEO/SEM, Brand Strategy, Content Marketing, and Data Analysis (SQL/BI).

Crucially, the analysis shows that for these specific matches, **0 training weeks are required**. This suggests that while there are skill overlaps, the system did not identify any placements that explicitly require reskilling weeks for Jake Peralta based on his current skill set and the available open positions.

Sample 08: Flight Risk Vacancy & Structural Safety Mitigation (Quiet Forces)

Prompt: "If Wanda Swanson leaves the company, perform a structural safety analysis. Execute the reallocation tool and ensure you pass an empty string for the source scope parameter. Find high-priority open positions company-wide."

Thought for 6 seconds

A structural safety analysis for Wanda Swanson's potential departure has been executed. The reallocation tool identified several potential matches for her skills across the company, primarily within the Marketing Department for a "Marketing Specialist" role.

Here's a summary of the findings:

- **Candidate:** Wanda Swanson
- **Current Department:** Finance
- **Target Department:** Marketing
- **Position Title:** Marketing Specialist
- **Hiring Manager:** Leia Organa
- **Skills Matched:** Project Management, Critical Thinking, Public Speaking, Corporate Taxation.
- **Required Skills for Target Role:** Growth Hacking, SEO/SEM, Brand Strategy, Content Marketing.
- **Training Weeks Required:** 0 (for the listed matches).

This indicates that Wanda Swanson possesses skills that are transferable to a Marketing Specialist role, with no additional training weeks required for the specific skill matches identified. This suggests a relatively low-risk transition in terms of skill gaps for these particular opportunities.

Sample 09: Cross-Departmental Bridge & Connection Weight Analysis

Prompt: "Execute a general organization discovery query to find employees bridging different departments. Specifically, list the connections and the relationship weights between individual nodes inside Engineering and nodes inside Sales."

Thought for 8 seconds

The analysis has identified several employees who bridge the Engineering and Sales departments, along with the collaboration weight between them.

Here are some examples of these cross-departmental connections:

- **Lee Beasley** (Engineering) collaborates with **Jim Halpert** (Sales) with a weight of **0.83**.
- **Alan Wallace** (Engineering) collaborates with **Kelly Lucas** (Sales) with a weight of **0.72**.
- **Terry Phillips** (Engineering) collaborates with **Jaime Horn** (Sales) with a weight of **0.90**.
- **Breanna Taylor** (Engineering) collaborates with **Joseph Miller** (Sales) with a weight of **0.70**.
- **Steven Hess** (Engineering) collaborates with **Matthew Lutz** (Sales) with a weight of **0.81**.
- **Andrew Lynch** (Engineering) collaborates with **Kathy Carter** (Sales) with a weight of **0.85**.
- **Eduardo Williams** (Engineering) collaborates with **Eric Mendez** (Sales) with a weight of **0.63**.
- **Travis Welch** (Engineering) collaborates with **Michelle Williams** (Sales) with a weight of **0.47**.
- **Helen Porter** (Engineering) collaborates with **Tyler Wright** (Sales) with a weight of **0.55**.
- **Amy Sloan** (Engineering) collaborates with **Dr. Laura Vang** (Sales) with a weight of **0.53**.

This list provides a detailed view of the collaborative relationships and their strengths between individuals in these two critical departments.

Sample 10: Hidden Mentor & Subject Matter Expert Identification (Python)

Prompt: "We have open positions requiring Python skills. Who are the highly proficient, high-influence Python experts inside the company who could act as upskilling mentors for internal transfers?"

Thought for 5 seconds

The following highly proficient and high-influence Python experts have been identified as potential upskilling mentors for internal transfers:

- **Miss Kimberly Clark** (IT Department) - Influence Score: 0.75
- **Samantha Sellers** (IT Department) - Influence Score: 0.87
- **Keith Smith** (IT Department) - Influence Score: 0.86
- **Jesse Mitchell** (IT Department) - Influence Score: 0.76

These individuals possess both high proficiency in Python and significant influence within the organization, making them ideal candidates to support internal talent development initiatives.

3. Conclusion

These inferences confirm that the NeoSmith agent consistently respects the structural constraints defined in our ADR and performs precise path traversals across the corporate grid.